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|------|------------------------------------------------------------------------------------------------------------|------|------------------------|-------------|
| 제품명  | 바크락스주250mg 설명서(수출)                                                                                         |      | 날짜                     | 수정진행사항      |
| 작업사양 | <div><div></div><div>인쇄도수 : 먹색1도</div><div>규격 : 110*205</div><div>K지질 : 70모조지</div><div>특이사항 :</div></div> |      | 2023. 12. 21           | 전체수정(데이터받음) |
| 출력일  | 2023. 12. 26                                                                                               | 도안확인 | <div><div></div></div> |             |

110\*205mm

Antiviral Agent

Vacrax\*

Injection 250mg

(Acyclovir)

**Presentation** : Vials containing the equivalent of 250mg sterile acyclovir as freeze dried sodium salts, a white to off-white powder

**Pharmacology**  
In adults, the terminal plasma half-life of acyclovir after administration of Vacrax I.V. is about 2.9 hours. Most of the drug is excreted unchanged by the kidney. Renal clearance of acyclovir is substantially greater than creatinine clearance, indicating that tubular secretion, in addition to glomerular filtration, contributes to the renal elimination of the drug. 9-carboxymethoxy-methylguanine is the only significant metabolite of acyclovir and accounts for 10 to 15% of the dose excreted in the urine.

When acyclovir is given one hour after 1 gram of probenecid, the terminal half-life and the area under the plasma concentration time curve, are extended by 18% and 40% respectively.

In adults, mean steady state peak plasma concentrations (C<sup>ss</sup>max) following a one-hour infusion of 2.5 mg/kg, 5 mg/kg and 10 mg/kg and 15mg/kg were 22.7 micromolar (5.1 microgram/ml), 43.6 micromolar (9.8 microgram/ml), 92 micromolar (20.7 microgram/ml) and 105micromolar(23.6microgram/ml), respectively. The corresponding trough levels (C<sup>ss</sup>min) 7 hours later were 2.2 micromolar (0.5 microgram/ml), 3.1 micromolar (0.7 microgram/ml), 10.2 micromolar (2.3 microgram/ml) and 808micromolar(2.0microgram/ml), respectively. In children over 1 year of age similar mean peak (C<sup>ss</sup>max) and trough (C<sup>ss</sup>min) levels were observed when a dose of 250 mg/m<sup>2</sup> was substituted for 5 mg/kg and a dose of 500 mg/m<sup>2</sup> was substituted for 10 mg/kg. In neonates (0 to 3 months of age) treated with doses of 10 mg/kg administered by infusion over a one-hour period every 8 hours the C<sup>ss</sup>max was found to be 61.2 micromolar (13.8 microgram/ml) and the C<sup>ss</sup>min to be 10.1 microgram/ml.

The terminal plasma half-life in these patients was 3.8 hours. In the elderly, total body clearance falls with increasing age and is associated with decreases in creatinine clearance although there is little change in the terminal plasma half-life.

In patients with chronic renal failure the mean terminal plasma half-life was found to be 19.5 hours. The mean acyclovir half-life during haemodialysis was 5.7 hours. Plasma acyclovir levels dropped approximately 60% during dialysis.

Cerebrospinal fluid levels are approximately 50% of corresponding plasma levels.

Plasma protein binding is relatively low (9 to 33%) and drug interactions involving binding site displacement are not anticipated.

**Indications** : Vacrax I.V. is indicated for the treatment of Herpes simplex infections in immunocompromised patients and severe initial genital herpes in the non-immunocompromised. Vacrax I.V. is indicated for the prophylaxis of Herpes simplex infections in immunocompromised patients. Vacrax I.V. is indicated for the treatment of Varicella zoster infections. Vacrax I.V. is indicated for the treatment of herpes encephalitis. Vacrax I.V. is indicated for the treatment of Herpes simplex infections in the neonate and infant up to 3 months of age.

**Dosage and administration:** A course of treatment with Vacrax I.V. usually lasts 5days, but this may be adjusted according to the patient's condition and response to therapy. Treatment for herpes encephalitis and neonatal Herpes simplex infections usually lasts 10 days. The duration of prophylactic administration of Vacrax I.V. is determined by the duration of the period at risk.

**Dosage in adults:** Patient with Herpes simplex (except herpes encephalitis) or Varicella zoster infections should be given Vacrax I.V. in doses of 5mg/kg bodyweight every 8 hours. Immunocompromised patients with Varicella zoster infections or patients with herpes encephalitis should be given Vacrax I.V. in doses of 10mg/kg bodyweight every 8hours provided renal function is not impaired. (see Dosage in renal impairment).

**Dosage in Children:** The dose of Vacrax I.V. for children aged between 3 months and 12 years is calculated on the basis of body surface area. Children with Herpes simplex(except herpes encephalitis) or Varicella zoster infection should be given Vacrax I.V. in doses of 250mg/m<sup>2</sup> of body surface area every 8 hours. In Immunocompromised children with Varicella zoster infections or children with herpes encephalitis, Vacrax I.V. should be given in doses of 500mg/ m<sup>2</sup> body surface area every 8 hours if renal function is not impaired. Children with impaired renal function require an appropriately modified dose according to the degree of impairment. The dosage of Vacrax I.V. in neonates and infants up to 3months of age is calculated on the basis of bodyweight. Neonates and infants up to 3months of age with Herpes simplex infections should be given Vacrax I.V. in doses of 10mg/kg bodyweight every 8hours. Treatment for neonatal herpes simplex infections usually lasts 10 days.

**Dosage in the elderly:** In the elderly, total acyclovir body clearance declines in parallel with creatinine clearance. Special attention should be given to dosage reduction in elderly patients with impaired creatinine clearance.

**Dosage in renal impairment:** Caution is advised when administering Vacrax Injection to patients with impaired renal function. The following adjustments in dosage are suggested;

| Creatinine Clearance Dosage |                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25 to 50 ml/min             | The dose recommended above(5 or 10mg/kg bodyweight or 500mg/m <sup>2</sup> ) should be given every 12 hours.                                                                                                                                                                                                                                                                               |
| 10 to 25 ml/min             | The dose recommended above(5 or 10mg/kg body weight or 500mg/m <sup>2</sup> ) should be given every 12 hours.                                                                                                                                                                                                                                                                              |
| 0(anuric) to 10ml/min       | In patients receiving continuous ambulatory peritoneal dialysis(CAPD) the dose recommended above(5 or 10mg/kg bodyweight or 500mg/m <sup>2</sup> ) should be halved and administered every 24 hours. In patients receiving haemodialysis the dose recommended above(5 or 10mg/kg bodyweight or 500mg/m <sup>2</sup> ) should be halved and administered every 24 hours and after dialysis. |

Reconstitution: Vacrax I.V. should be reconstituted using the following volumes of either Water for Injection BP or Sodium Chloride Intravenous Injection BP(0.9% w/v) to provide a solution containing 25mg acyclovir per ml ;

|             |                                    |
|-------------|------------------------------------|
| Formulation | Volume of fluid for reconstitution |
| 250mg vial  | 10mL                               |

From the calculated dose, determine the appropriate number and strength of vials to be used. To reconstitute each vial add the recommended volume of infusion fluid and shake gently until the contents of the vial have dissolved completely.

**Administration:** The required dose of Vacrax I.V. should be administered by slow intravenous infusion over a one-hour period. After reconstitution Vacrax I.V. may be administered by a controlled-rate infusion pump. Alternatively, the reconstituted solution may be further diluted to give an acyclovir concentration of not greater than 5mg/ml(0.5% w/v) for administration by infusion: Add the required volume of reconstituted solution to the chosen infusion solution, as recommended below, and shake well to ensure adequate mixing occurs. For children and neonates, where it is advisable to keep the volume of infusion fluid to a minimum, it is recommended that dilution is on the basis of 4ml reconstituted solution (100mg acyclovir) added to 20ml of infusion fluid.

For adults, it is recommended that infusion bags containing 100ml of infusion fluid are used, even when this would give an acyclovir concentration substantially below 0.5% w/v. Thus one 100ml infusion bag may be used for any dose between 250mg and 500mg acyclovir(10 and 20ml of reconstituted solution) but a second bag must be used for doses between 500 and 1000mg. When diluted in accordance with the recommended schedules, Vacrax I.V. is known to be compatible with the following infusion fluids and stable for up to 12 hours at room temperature(15°C to 25°C) ; Sodium Chloride Intravenous Infusion BP(0.45% and 0.9% w/v); Sodium Chloride(0.18% w/v) and Glucose(4% w/v) Intravenous Infusion BP; Sodium Chloride(0.45% w/v) and Glucose(2.5% w/v) Intravenous Infusion BP; Compound Sodium Lactate Intravenous Infusion BP(Hartmann's Solution). Vacrax Injection when diluted in accordance with the above schedule will give an acyclovir concentration not greater than 0.5% w/v. Since no antimicrobial preservative is included, reconstitution and dilution must be carried out under full aseptic conditions, immediately before use and any unused solution discarded. Should any visible turbidity or crystallization appear in the solution before or during infusion, the preparation should be discarded

**Contraindications, Warnings, etc**

**Contraindication:** Vacrax I.V. is contraindicated in patients known to be previously hypersensitive to acyclovir.

**Precaution:** The dose of Vacrax I.V. must be adjusted in patients with impaired renal function in order to avoid accumulation of acyclovir in the body(see Dosage in renal impairment). In patients receiving Vacrax I.V. at higher doses(eg. For herpes encephalitis), specific care regarding renal function should be taken, particularly when patients are dehydrated or have any renal impairment. Reconstituted Vacrax I.V. has a pH of Approximately 11.0 and should not be administered by mouth. The results of a wide range of mutagenicity test in vitro and in vivo indicate that acyclovir is unlikely to, pose a generic risk to man.

Acyclovir was not found to be carcinogenic in long term studies in the rat and the mouse. Systemic administration of acyclovir in internationally accepted standard tests did not product embryotoxic or teratogenic effects in rabbits, rats or mice. In a non-standard test in rats, foetal abnormalities were observed but only following such high subcutaneous doses that maternal toxicity was produced. The clinical relevance of the these findings is uncertain. Largely reversible adverse effects on spermatogenesis in association with overall toxicity in rats and dogs have been reported only at doses of acyclovir greatly in excess of those employed therapeutically. Two-generation studies in mice did not reveal any effect of (orally administered) acyclovir in fertility. There is no experience of the effect of Vacrax I.V. on human fertility.

Vacrax tablets have been shown to have no definitive effect upon sperm count, morphology or motility in man.

**Side and adverse effects:** Rapid increases in blood urea and creatinine levels may occasionally occur in patients given Vacrax I.V. This is believed to be related to peak plasma levels and the state of hydration of the patients. To avoid this effect the drug should not be given as an intravenous bolus injection but by slow infusion over a one hour period. Adequate hydration of the patient should be maintained. Renal impairment developing during treatment with Vacrax I.V. usually responds rapidly to rehydration of the patient and/or dosage reduction or withdrawal of the drug.

Progression to acute renal failure, however, can occur in exceptional cases. Severe local inflammatory reactions sometimes leading to breakdown of the skin have occurred when Vacrax I.V. has been inadvertently infused into extravascular tissues. Reversible neurological reactions such as confusion, hallucinations, agitation, tremors, somnolence, psychosis, convulsions and coma have been associated with Vacrax I.V. therapy, usually in medically complicated cases. Nausea and vomiting have been reported in patients receiving therapy with Vacrax I.V. Other events reported in patients receiving Vacrax I.V. include increases liver-related enzymes, rashes and fevers and decrease in haematological indices (anaemia, thrombocytopenia, leucopenia).

**Use in pregnancy and lactation:** Limited data are available on the use of acyclovir during pregnancy. Caution should therefore be exercised by balancing the potential benefits of treatment against any possible hazard. Following oral administration of 200mg five times a day, acyclovir has been detected in human breast milk at concentrations ranging from 0.6 to 4.1 times the corresponding plasma levels. These levels would potentially expose nursing infants to acyclovir dosages of up to 0.3mg/kg bodyweight/day. Caution is therefore advised if Vacrax Injection is to be administered to a nursing woman.

**Drug interactions:** Probenecid increases the acyclovir mean half-life and area under the plasma concentration time curve. Other drugs affecting renal physiology could potentially influence the pharmacokinetics of acyclovir. However, clinical experience has not identified other drug interactions with acyclovir.

**Toxicity and treatment of overdose:** Single doses of Vacrax I.V. up to 80mg/kg bodyweight have been inadvertently administered without adverse effects. Acyclovir is dialyzable.

**Incompatibilities:** None reported so far

**Storage condition:**  
Store below 25°C  
Vacrax I.V. contains no antimicrobial preservative. Reconstitution or dilution should therefore be carried out under full aseptic conditions immediately before use and any unused solution discarded. Reconstituted or diluted solutions should not be refrigerated.

**How supplied:**  
Vacrax™ 250mg ; pack of 10vials

Keep dry and protect from light.

☞ Keep out of the reach of children

\*Revised Date : April 2026

**Manufactured by:**  
**A PROGEN**  
Pharmaceuticals, Inc.  
16, Dumeori-gil, Yanggam-myeon, Hwaseong-si, Gyeonggi-do, Republic of Korea.

**Product Registration Holder:**  
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